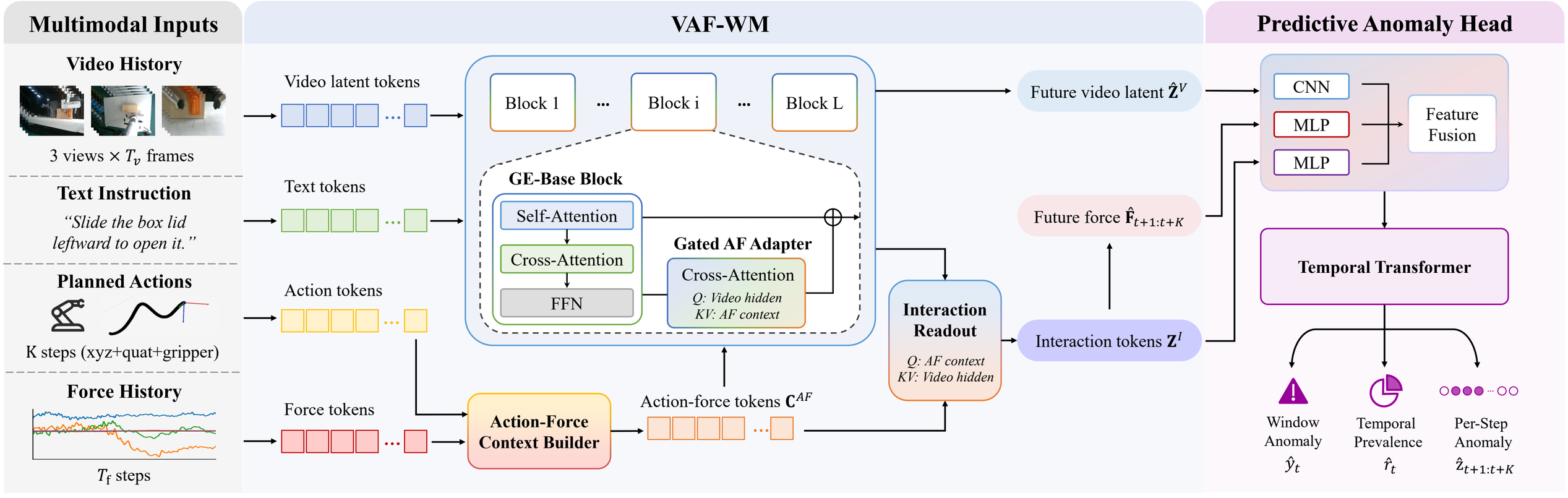

Multimodal Inputs
Video History
3 views × T_v frames
Text Instruction
"Slide the box lid leftward to open it."
Planned Actions
K steps (xyz+quat+gripper)
Force History
T_f steps
VAF-WM
Video latent tokens
Text tokens
Action tokens
Force tokens
Block 1
Block i
Block L
GE-Base Block
Self-Attention
Cross-Attention
FFN
Gated AF Adapter
Cross-Attention
Q: Video hidden
KV: AF context
Action-Force Context Builder
Action-force tokens $\mathbf{C}^{AF}$
Interaction Readout
Q: AF context
KV: Video hidden
Future video latent $\hat{\mathbf{Z}}^V$
Future force $\hat{\mathbf{F}}_{t+1:t+K}$
Interaction tokens $\mathbf{Z}^I$
Predictive Anomaly Head
CNN
MLP
MLP
Feature Fusion
Temporal Transformer
Window Anomaly $\hat{y}_t$
Temporal Prevalence $\hat{r}_t$
Per-Step Anomaly $\hat{\mathbf{z}}_{t+1:t+K}$